

KHOLOUD NASSER

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SUMMARY

Experienced software developer with over 11+ years of expertise in full-stack development using C#, .NET, Java, Node.js and modern web technologies. Proven ability in designing scalable and secure software architectures, leading cross-functional development teams, and delivering high-performance applications for internal and client-facing platforms. Skilled in crafting robust backend systems, developing REST APIs, and optimizing system performance and availability. Adept at aligning technical solutions with business goals while mentoring team members to ensure code quality, agility, and delivery excellence.

EXPERIENCE

Senior Software Developer

December 2013 - November 2025

- **Emirates Red Crescent | Abu Dhabi**
 - Developed and maintained high-performance backend systems using C# and .NET, resulting in a 30% increase in system efficiency and reducing response times by 50% for both internal and client-facing applications.
 - Ensured system security and scalability, implementing best practices that reduced security breaches by 95% and supported a 200% increase in user traffic without performance degradation.
 - Design and implement RESTful APIs with .NET Core, optimizing data handling, authentication, and real-time communication across multiple platforms.
 - Utilized ASP.NET MVC and WebAPI to build user-friendly web applications with responsive UI/UX components, improving user engagement by 30% and ensuring seamless functionality across multiple devices and screen sizes.
 - Troubleshoot and resolved performance bottlenecks, security vulnerabilities, and application bugs within the .NET ecosystem, enhancing system stability and reducing downtime to 0.1%, ensuring 99.9% uptime across critical applications.
 - Lead the development of engaging game features, including designing interactive systems and user experiences for mobile games, focusing on player engagement and real-time interaction with AI-driven characters.
 - Implemented donations through MPGS (Mastercard Payment Gateway Services), enabling secure and seamless transactions, which increased donation processing efficiency by 40% and reduced transaction errors by 15%.

PROJECT

Academic Project – VR & Eye Tracking System for ADHD Intervention

- **Heriot-Watt University**
 - Designed and developed a **multi-modal VR application** using *Unity3D* and the **Meta Quest Pro** headset to assess and support attention in individuals with ADHD.
 - Implemented **real-time gaze tracking** and **adaptive difficulty logic** to adjust tasks dynamically based on attention lapses.
 - Built a secure **Azure backend** with Function Apps and SQL Database for session logging, analytics, and performance metrics.
 - Developed a **web-based dashboard (ASP.NET Core, Razor Pages)** to visualize task results, gaze data, and adaptive changes for researchers and clinicians.

- Conducted a **pilot study with 10 participants**, analyzing reaction times, gaze fixation, and distractor effects. Results showed clear differentiation between focused vs. distracted performance.

Academic Project – Image Classification Using ML & Deep Learning Models

- **Heriot-Watt University**
 - Conducted a comparative study between traditional Machine Learning models (Random Forest, Linear Classifier, Perceptron) and Convolutional Neural Networks (CNNs) for an image classification dataset.
 - Implemented and evaluated multiple models including 10-fold Stratified Cross Validation, MLP, CNN, and ANN, using various activation functions (ReLU, Tanh, Sigmoid).
 - Achieved ~92% accuracy using Perceptron and ~99.9% with CNNs under specific conditions.
 - Investigated model performance, generalization ability, and convergence with a focus on hyperparameter tuning and overfitting control.

EDUCATION

Master's of Science in Software Engineering

- **Heriot Watt University**
Dubai, United Arab Emirates • 2025 • 4.00 GPA

Bachelor's of Computers and Information

Minor in Information Technology

- **Assiut University**
Assiut, Egypt • 2013
 - **Graduation Project:** Network Traffic Monitoring Application developed in Java, featuring MySQL database integration and GUI-based analytics.

CERTIFICATIONS

ITIL Foundation in IT Service Management

2017

- **APMG International Accrediting Professionals**

PUBLICATIONS

2025

Paper

- **AAAI-SuS-2025** | The Design of a Virtual Reality Environment for ADHD Intervention

SKILLS

Programming Languages: C#, C++, Java, JavaScript, Node.js, Python, PHP

Java Technologies: Java SE/EE, Spring Framework (Spring Boot, Spring MVC), Maven, Gradle, JUnit

Backend Technologies: .NET Core, ASP.NET, Entity Framework, REST APIs, SOAP

Frontend Technologies: Html, CSS, JavaScript, React

Game Development: Unity, WebGPU, Unreal Engine

Databases: Oracle, SQL Server, MongoDB, MySQL

DevOps & Cloud Tools: Microsoft Azure, Docker, Kubernetes, Jenkins, Git, Jira